TEXORA · LECTURE SHEET

Warping Calculations

Warping turns hundreds of cones into one ordered sheet of parallel ends on a beam. Its calculations are bookkeeping with consequences: ends per beam against creel capacity, beam weight, beam capacity, winding time, and the number of beams in a set — the arithmetic that Sheet 6's whole-shed project plan is built from.

FM-3207 · ADVANCED WEAVING AND KNITTING CALCULATIONS · TOPIC 2 · SPECIALIZATION — FABRIC MANUFACTURING

1. Ends, creel and the number of beams

A warp of E total ends is wound from a creel holding C packages. On a direct (beam) warper one beam takes at most C ends; on a sectional warper the warp is built band by band to the same arithmetic. The number of beams (or sections) is therefore

$$\text{Beams per set} = \text{total ends} \div \text{creel capacity, rounded UP}$$

WORKED EXAMPLE — BEAMS IN A SET

A fabric needs **4,200 total ends**; the warping creel holds **560 cones**.

Beams = 4,200 ÷ 560 = 7.5 → **8 beams**: seven beams of 560 ends and one beam of 4,200 − 7 × 560 = **280 ends**.

The short beam is not an inconvenience to be hidden: at sizing the eight beams are combined, and the sizing plan (Sheet 3) must know that one beam carries half the tension load of the others, or the sheet will not run level.

2. Beam weight and winding time

$$\text{Yarn on one beam (kg)} = \text{ends} \times \text{length (m)} \times \text{tex} \div 1{,}000{,}000$$

$$\text{Winding time} = \text{beam length} \div (\text{warping speed} \times \text{efficiency})$$

WORKED EXAMPLE — ONE WARPER'S BEAM

560 ends, beam length 6,000 m, yarn Ne 20 (29.53 tex), warping speed 800 m/min, efficiency 70%.

Beam weight = 560 × 6,000 × 29.53 ÷ 10^6 = **99.2 kg**.

Winding time = 6,000 ÷ (800 × 0.70) = **10.7 min** of running — but a beam change, creel check and a single end-break stoppage are all inside the 70%, which is why the efficiency, not the catalogue speed, belongs in the plan.

A 40-beam programme on this duty consumes 40 × 99.2 ≈ **3,968 kg** of warp yarn — the purchase figure before sizing add-on is even considered.

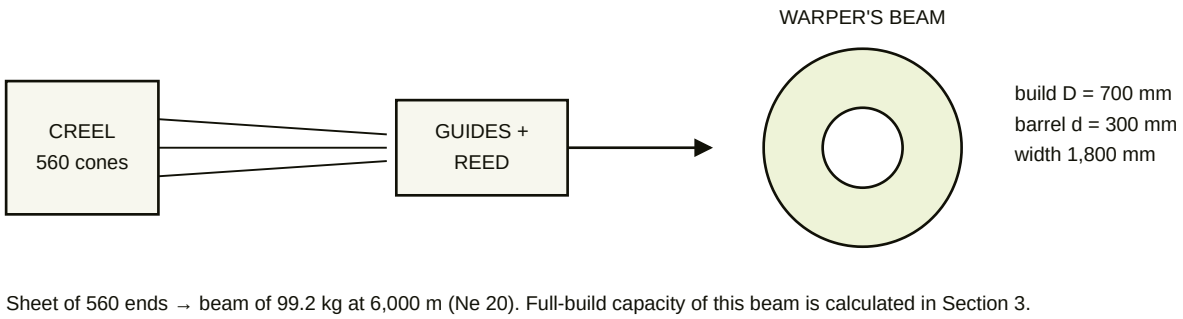


Figure 1 — Warping line: creel, guide/reed, beam — with the beam build dimensions used in Section 3.

3. Beam capacity — how much yarn the beam can hold

The yarn on a full beam occupies the annulus between barrel and build diameter, across the beam width, at the winding density of a parallel sheet under tension (about 0.45 g/cm³ for cotton warp):

$$\text{Beam mass} = (\pi/4) \times (D^2 - d^2) \times \text{width} \times \text{density}$$

WORKED EXAMPLE — FULL-BUILD CAPACITY

D = 700 mm, d = 300 mm, width 1,800 mm, density 0.45 g/cm³.

Volume = (π/4) × (70² − 30²) × 180 = (π/4) × 4,000 × 180 ≈ 565,500 cm³.

Mass = 565,500 × 0.45 ≈ **254.5 kg**.

At 560 ends of 29.53 tex (16.54 g per metre of sheet), that mass is 254,500 ÷ 16.54 ≈ **15,390 m** of warp — the theoretical ceiling. Practical beam lengths are set well below it: flange strength, sizing-machine creel handling and the loom's beam change rhythm all bite first, which is why the 6,000 m beam of Section 2, not the 15,390 m ceiling, is the planning figure.

4. Sectional warping arithmetic

On a sectional warper the same totals are reached band by band: ends per section = creel capacity; section width = fabric (reed) width × (section ends ÷ total ends); the drum is traversed by exactly one section width per section so the bands tile the full width without gaps or overlaps. For the 4,200-end warp on a 560-cone creel with a reed width of 1.90 m: 8 sections, each 560 ends except the last (280),

and each full section is 1.90 × 560/4,200 = 0.253 m wide. The cone angle of the drum compensates the build so every section finishes at the same diameter — the condition for even tension when the sections are beamed off together onto the weaver's beam.

Glossary

Creel capacity — number of supply packages the warping creel holds; the maximum ends per beam or section. **Set** — the group of warper's beams that combine at sizing into one weaver's beam. **Beam build** — the outside diameter of yarn wound on a beam. **Section** — one band of a sectional warp. **Beaming off** — transferring the sections from the warping drum to the weaver's beam.

Key points

• Beams per set = total ends ÷ creel capacity, rounded up (worked: 4,200/560 → 8 beams, last of 280 ends).
• Beam weight = ends × length × tex ÷ 10^6 (worked: 99.2 kg); winding time uses efficiency, not catalogue speed (10.7 min).
• Beam capacity from the annulus volume × density is a ceiling (254.5 kg ≈ 15,390 m), not a plan.
• Sectional warping tiles the reed width: section width = reed width × section ends/total ends.

Practice

1. A warp of 5,280 ends is prepared on a 528-cone creel. How many beams are in the set, and are they all full?
2. Find the yarn weight of a beam of 528 ends, 2,725 m, Ne 24 (24.6 tex).
3. A sectional warper winds the 5,280-end warp to a reed width of 1.524 m (60 in). What is the width of each full section?
4. Why is beam capacity (Section 3) never used directly as the beam length in a production plan?

Answer key

1. 5,280 ÷ 528 = **10 beams exactly — all full**, no short beam. (This is the shed of Sheet 6, whose creel was chosen to divide the warp exactly.)
2. 528 × 2,725 × 24.6 ÷ 10^6 = **35.4 kg**.
3. Section width = 1.524 × 528/5,280 = **0.1524 m (152.4 mm)** per section; ten sections tile the 60-inch reed width exactly.
4. Capacity is a geometric ceiling at full build and perfect density. Real beam length is limited earlier by flange strength and handling weight, by how much warp the sizing creel and the loom's beam

gate can manage, and by matching beam changes to the production rhythm — a beam that technically fits can still be the wrong beam.
